
CS246: Database Management Systems Lab

Week # 07 (1 Questions, 159 Marks)

Lab session: AL1

Held on: 17-Feb-2025 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 4

Submission time: 16:45 Hrs, 17-Feb-2025

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Important information:

1. Some of the tasks are so designed to emit errors when SQL statements are invoked. You should make an informed effort to find the cause of the errors. You do not need to resolve the errors.
2. Write all the SQL statements in a text file named with `your-roll-number.sql` file name & extension and upload. Replace the text `your-roll-number` with appropriate roll number.
3. Given the database, generate several `insert` statements using a C program. Execute the
4. Refer to table creation document
5. Refer to insert statement
6. Refer to select statement
7. Refer to join statement
8. Refer to load statement

Question 1: (159 points)

Go through each of the described task and develop SQL queries. These SQL queries are based on contents

- Section 4.1 (join)
- Section 3.7 (Aggregate functions)
- Section 3.8 (Nested queries)

Basic SQL, Chapter 3 and Chapter 4, text book *Database System Concepts*, Abraham Silberschatz, Henry F. Korth & S. Sudarshan.

Task 01 (1 mark) Create a database named `week07`. Create the following tables within this database.

Task 02 (28 marks) Create the following tables

1. (5 marks) Table named **myapp_user** with the following schema

1 st column	user_id	UNIQUE, PRIMARY KEY	int data type
2 nd column	name	NOT NULL	fixed length char of size 100
3 rd column	email	UNIQUE, NOT NULL	fixed length char of size 100
4 th column	phone	UNIQUE, NOT NULL	fixed length char of size 15
5 th column	city	NOT NULL	fixed length char of size 50

2. (4 marks) Table named **restaurant** with the following schema

1 st column	restaurant_id	PRIMARY KEY	int data type
2 nd column	name	NOT NULL	fixed length char of size 100
3 rd column	location	NOT NULL	fixed length char of size 100
4 th column	rating	NOT NULL	DECIMAL(3, 2)

3. (7 marks) Table named **order** with the following schema

1 st column	order_id	PRIMARY KEY	int data type
2 nd column	user_id	NOT NULL	int data type
3 rd column	restaurant_id	NOT NULL	int data type
4 th column	order_date	NOT NULL	DATE
5 th column	amount	NOT NULL	DECIMAL(10, 2)

(1 mark) **user_id** foreign key referencing **myapp_user** table; Specify cascade on delete

(1 mark) **restaurant_id** foreign key referencing **restaurant** table; Specify cascade on delete

4. (5 marks) Table named **menu** with the following schema

1 st column	menu_id	PRIMARY KEY	int data type
2 nd column	restaurant_id	NOT NULL	int data type
3 rd column	dish_name	NOT NULL	fixed length char of size 100
4 th column	price	NOT NULL	decimal(10, 2)

(1 mark) **restaurant_id** foreign key referencing **restaurant** table; Specify cascade on delete

5. (7 marks) Table named **review** with the following schema

1 st column	review_id	PRIMARY KEY	int data type
2 nd column	user_id	NOT NULL	int data type
3 rd column	restaurant_id	NOT NULL	int data type
4 th column	rating	NOT NULL	decimal(3, 2)
5 th column	review_text	NOT NULL	TEXT

(1 mark) `user_id` foreign key referencing `myapp_user` table; Specify cascade on delete

(1 mark) `restaurant_id` foreign key referencing `restaurant` table; Specify cascade on delete

Task 03 (10 marks) Loading data.

1. (2 marks) Use the `load` statement to load `users.csv` file into `myapp_user` table
2. (2 marks) Use the `load` statement to load `restaurants.csv` file into `restaurant` table
3. (2 marks) Use the `load` statement to load `menu_items.csv` file into `menu` table
4. (2 marks) Use the `load` statement to load `orders.csv` file into `myorder` table
5. (2 marks) Use the `load` statement to load `reviews.csv` file into `review` table

Task 04 (120 marks) Write SQL statements for the following queries.

1. (20 marks) Queries involving two tables
 - (a) (5 marks) Obtain user name, order date and order amount
 - (b) (5 marks) Obtain restaurant name, dish name and price
 - (c) (5 marks) Obtain user name, rating, review text
 - (d) (5 marks) Obtain user name, restaurant name and order amount
2. (20 marks) Queries involving three tables
 - (a) (5 marks) Obtain user name, dish name and price ordered by user
 - (b) (5 marks) Obtain user name, restaurant name, rating and text comment given by user
 - (c) (5 marks) Obtain user name and total order amount day wise
 - (d) (5 marks) Find the name of the user who made maximum orders in a month
3. (20 marks) Queries involving two tables
 - (a) (5 marks) Find all users (retrieve user name) who never placed an order
 - (b) (5 marks) Find all restaurants (retrieve restaurant name) that do not have any menu item
 - (c) (5 marks) Find restaurants that never received any order (retrieve restaurant id and restaurant name)
 - (d) (5 marks) Find all users (retrieve user name) who never written a review
4. (20 marks) Queries involving aggregate functions
 - (a) (5 marks) Find the total number of orders placed
 - (b) (5 marks) Find the restaurant with the minimum order amount (retrieve restaurant name and amount columns)
 - (c) (5 marks) Find the restaurant with the maximum rating from reviews (retrieve restaurant name and rating columns)

- (d) (5 marks) Find the average order amount for each restaurant (retrieve restaurant name and amount columns)
- 5. (40 marks) Write nested queries for the following
 - (a) (10 marks) Find users who have placed the most orders (retrieve user id, user name)
 - (b) (10 marks) Find the restaurants with the highest average rating (retrieve restaurant id and restaurant name)
 - (c) (10 marks) Find users who have spent more than the average order amount across all users (retrieve user id, user name)
 - (d) (10 marks) Find restaurants that have never been reviewed (retrieve restaurant id and restaurant name)